

ML-UNG: Prefix-Frontier Entries and Predicate-Safe Block Navigation for Filtered Vector Search

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ABSTRACT

Label-filtered approximate nearest-neighbor search must connect eligible vectors without letting metadata processing dominate search. In a containment-based index, replacing dense group relations by a prefix tree changes which entry groups are required; retaining the original entries can leave eligible branches unreachable. We present ML-UNG, which couples a terminal-prefix forest with a bitmap-backed covering frontier, and introduces independently constructed blocks for coarser predicate-safe navigation. A block is authorized when its root prefix contains the query labels. Search states carry a physical level and expand only that level’s edges. We prove label-plane coverage and conditional predicate safety, while treating finite-budget ANN Recall as an experimental property. A structural rule instantiates overlay count and thresholds without query-performance calibration, and batched GPU tasks accelerate construction. On Amazon, matched 0L Trie/entry search is 2.77–18.43× faster than the optimized LNG system at three measured selectivities below 6%, but is slower or lacks a Recall crossing on broader workloads. The automatic LNG-base plan, 2L[L|LT]/DRH-v1, achieves 14.40–26.70× over optimized 0L[L] at 60–99% selectivity. These are screening results at measured Recall@10 crossings; held-out regressions remain. The fastest composed GPU-assisted build takes 141.27 seconds versus 190.38 seconds for the original CPU base, with query-quality equivalence across construction backends still to be established.

1 INTRODUCTION

Vector search increasingly operates inside a structured database query. A product search combines semantic similarity with category and availability constraints; a retrieval application restricts documents by topic, language, or access labels before ranking their embeddings. In each case the answer must be close to a query vector and satisfy the metadata predicate. This paper studies conjunctions of labels: a vector is eligible when its label set contains every query label. We define selectivity as the fraction of eligible vectors, so a low-selectivity query admits a small fraction of the database.

The difficulty is that predicate eligibility and geometric proximity induce different organizations of the same data. An unfiltered proximity graph can place eligible vectors behind many ineligible neighbors. Filter-aware search and construction, including ACORN [11], address this tension through richer neighborhoods and query-time expansion. Label-aware indexes instead organize navigation around metadata. UNG [2] connects exact-label groups through set containment, while Curator [6] shares label-specialized search structures through a clustering tree. These approaches offer different ways to recover useful navigation under a filter; selectivity alone does not determine which one is fastest.

We focus on a remaining coupling between *entry discovery* and *navigation granularity*. UNG starts from the minimal observed supersets of the query labels, then follows containment-directed cross-group edges. Computing this minimal set can require comparing eligible label sets across unrelated prefix branches. Once the entries are found, vectors are still partitioned by exact label equality, even when a query admits many groups that could be searched together. In our Amazon index, 602,453 vectors occupy 482,387 exact groups. This fragmentation makes entry work and cross-group navigation separate costs rather than interchangeable measures of index quality.

Our measurements illustrate both sides of the coupling. At 0.499% selectivity, even the optimized LNG-semantic provider takes 20.352 ms/query in a separate instrumented run, compared with 0.195 ms for vector-entry setup and 2.076 ms for graph traversal. A prefix-based system reduces discovery to 0.645 ms, but at 9.907% selectivity its graph traversal rises from the LNG system’s 98.747 ms to 323.081 ms. Faster discovery does not imply faster search: changing the entry contract changes the initial frontier and the geometric paths that must be explored. Figure 2 presents this tradeoff rather than attributing every difference to one component.

Our first idea is to replace global minimal-superset reduction with a *prefix frontier*. Fix a canonical order of labels and form a trie of observed label sets. Every eligible terminal belongs to a branch below a first eligible terminal. Seeding those first terminals covers every eligible group in the terminal-prefix forest. Thus a prefix topology and its frontier can be designed together: the entry computation need not discard a terminal merely because another branch contains a smaller label set. This removes a specific minimality computation, while retaining real costs for bitmap intersection, ancestor checks, and vector seed setup. The frontier can also be larger than UNG’s entry set; that tradeoff is part of the design.

Our second idea is to search *predicate-certified blocks* instead of always paying for exact-group boundaries. A block rooted at trie prefix R_b contains direct members whose label sets all include R_b . When $Q \subseteq R_b$, the query may traverse a local vector graph spanning these groups without admitting an ineligible vector. A finer representation is still necessary: a query containing a label absent from R_b cannot certify the whole block, even though some members may satisfy it. We therefore retain the exact-group base and independently materialize coarser block overlays. This yields a physical level-0/level-1 design as the simplest instance and extends to further scales without changing finer partitions.

ML-UNG combines these two ideas with explicit search ownership. Each candidate carries a physical level and expands only adjacency belonging to that level. Each level may use prefix or LNG connectivity; the choice is empirical because sparse label connectivity alone does not guarantee good vector navigation. Block certification establishes predicate safety, while frontier coverage is

a group-level property. Neither property promises exact nearest neighbors from a finite search budget. This distinction lets us explain both gains and regressions without hiding them behind a correctness claim.

The system must also be practical to build and configure. We batch the many irregular within-block and cross-block construction tasks on the GPU, retaining explicit memory and host-output boundaries. A secondary Degree-Ratio Hierarchy (DRH) rule derives overlay thresholds and depth from dataset cardinality and existing degree budgets without query-performance calibration. Its held-out failures are reported alongside its favorable cases; it is a reproducible heuristic rather than a runtime-optimality theorem.

This paper makes three contributions:

- **A prefix-frontier entry contract.** We define the entry set for terminal-prefix connectivity, show group-level coverage, and explain the cost and coverage differences from minimal-superset entry discovery.
- **Certified block navigation.** We construct independent label-prefix blocks and give a level-local query algorithm with explicit eligibility, seed, and fallback conditions. The complete base-by-overlay factorial exposes when sparse topology and coarser scales help or hurt.
- **Construction and empirical design guidance.** We provide a batched GPU construction pipeline and a query-calibration-free hierarchy rule, and evaluate them separately from topology and entry choices.

At measured Recall@10 ≥ 0.90 , the matched-provider Trie base is 2.77–18.43 $\times$ faster than the LNG base at the three Amazon workloads near 0.5–5% selectivity. Among the 14 evaluated topology configurations, Trie-base systems win seven of nine workloads; 0L LNG wins near 10% and 30%. The fastest measured GPU-assisted construction composition takes 141.27 s versus 190.38 s for the original CPU base builder. These are scoped screening and construction results, not a claim of superiority over all filtered-search systems or all predicates.

2 PROBLEM AND MOTIVATION

2.1 Label-Constrained Nearest Neighbors

Let $\mathcal{D} = \{(x_i, A_i)\}_{i=1}^N$, with $x_i \in \mathbb{R}^d$ and finite, nonempty observed label sets A_i (the scope accepted by the current Trie implementation). Query (q, Q, K) asks for the K nearest vectors in

$$\mathcal{E}(Q) = \{i : Q \subseteq A_i\}, \quad s(Q) = |\mathcal{E}(Q)|/N. \quad (1)$$

The distance function is fixed across compared methods. For workloads with at least K eligible points, Recall@ K is the intersection of the returned and exact filtered top- K sets, divided by K . If fewer than K points are eligible, an implementation must instead return at most that number and use that eligible count as the denominator; an empty eligible set has no ANN ranking task. The measured workloads use $K = 10$ and exact containment ground truth. Query $Q = \emptyset$ is valid and admits all vectors.

Every distinct observed label set S defines an exact group $G_S = \{i : A_i = S\}$. Let G denote the number of such groups, distinct from the vector count N . Labels are unique within a set and sorted under one fixed total order. The canonical label trie stores this sequence;

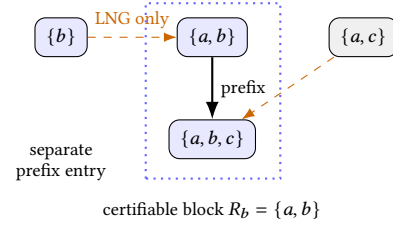


Figure 1: Illustrative label relations for $Q = \{b\}$. Blue groups are eligible; the gray group is not. The prefix frontier contains both top blue groups. Dashed LNG relations and the dotted block boundary explain different mechanisms; the drawing is not a measured vector layout.

its terminals represent observed groups. A group-level edge describes permitted navigation between groups. Materializing it as vector edges is a separate construction step, so group-edge counts are not vector-edge counts.

2.2 An Application Example

Consider a catalog in which a denotes English, b denotes books, and c denotes in-stock. A semantic query constrained to books has $Q = \{b\}$. Groups $\{b\}$, $\{a, b\}$, and $\{a, b, c\}$ are eligible; group $\{a, c\}$ is not. Under order $a < b < c$, the first two eligible groups lie on different prefix branches. UNG’s minimal-superset entry set contains only $\{b\}$, because set-containment edges can reach the larger groups. A prefix graph needs both $\{b\}$ and $\{a, b\}$ as entries. Removing the second entry would remove an entire eligible branch from graph-level coverage.

Conversely, a block rooted at prefix $\{a, b\}$ may safely join the $\{a, b\}$ and $\{a, b, c\}$ groups for query $\{b\}$. Query $\{b, c\}$ cannot certify that same block; it must use a finer representation containing the eligible $\{a, b, c\}$ group. This is the reason for two granularities: block size trades geometric freedom against the strength of the predicate certificate, rather than permitting filter violations inside large blocks.

2.3 Where Existing Organizations Spend Work

A global proximity graph, a label-partitioned index, a containment graph, and a filter-first scan pay for different work. Figure 2 summarizes these organizations. It is not a ranking of implementations. With a global graph, sparse eligible vertices can force extra traversal or neighborhood expansion; richer construction may alleviate the problem. With label-specialized partitions, queries must identify and combine suitable partitions. Curator’s clustering-based virtual indexes address this explicitly [6]; a temporary index is not evidence of a large distance-computation cost by itself. With UNG, entry discovery and fine-group traversal both remain relevant. Filter-first exact scanning has $\Theta(|\mathcal{E}(Q)|d)$ distance work even when the fraction $s(Q)$ is small, because a large database can still yield many eligible vectors.

UNG’s LNG is the cover graph of observed strict set containment: $S \rightarrow S'$ if $S \subset S'$ and no observed label set is strictly between them. It uses a trie to discover minimal observed supersets;

Global proximity graph Query $\rightarrow$ filtered expansion Work: examined neighborhoods Tension: eligible connectivity	Label-aware partitions Query $\rightarrow$ partition traversal Work: filter preparation + search Tension: composition and pruning	Containment graph Query $\rightarrow$ minimal supersets Work: entry + exact-group search Tension: many fine boundaries	Filter-first scan Query $\rightarrow$ eligible IDs $\rightarrow$ scan Work: $ \mathcal{E}(Q) $ distances Tension: absolute candidate mass
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	Mean selectivity	0L system	Discovery (ms)	Seed setup (ms)	Graph (ms)
0.499%		LNG + optimized LNG entry	20.352	0.195	2.076
		Trie + prefix-frontier entry	0.645	1.769	0.868
9.907%		LNG + optimized LNG entry	36.631	0.646	98.747
		Trie + prefix-frontier entry	3.778	14.744	323.081

Figure 2: Top: conceptual sources of work, not measured shortcomings of every system in each class. Bottom: measured per-query stage times from the separate Amazon profile at each system’s Recall crossing. Both bases use their matched entry provider, so this is a system comparison. Timings include instrumentation and concurrency effects; they are not the batch latency used to compute primary QPS. The measured rows quantify only the two matched 0L systems.

that lookup trie does not define its cross-group edges. Our prefix instance changes both the group relation and the matching entry contract. Our block overlays then change the navigation unit. These interventions must be measured separately: neither a faster entry routine nor fewer group edges alone establishes a faster filtered nearest-neighbor index.

3 PREFIX-FRONTIER ENTRY DISCOVERY

3.1 Entry Coverage Without Minimal-Superset Reduction

Let $C_Q = \{S : Q \subseteq S, G_S \neq \emptyset\}$ be the eligible observed label sets. Write $A <_p S$ when the sorted sequence of A is a proper prefix of that of S . Our entry frontier is

$$F(Q) = \{S \in C_Q : \nexists A \in C_Q \text{ with } A <_p S\}. \quad (2)$$

UNG’s entry set instead excludes S whenever an eligible proper subset of S is observed. Every prefix is a subset, but many subsets are not prefixes. Consequently the LNG-minimal entry set is contained in $F(Q)$, and the containment can be strict. Figure 1 shows the smallest relevant distinction: $\{b\} \subset \{a, b\}$ is not a prefix relation under $a < b$, so both groups must remain in $F(\{b\})$.

Connect each terminal to its nearest terminal descendants in the canonical trie. This forms a directed forest, with one incoming edge per non-root terminal. It has exactly $G - r$ group edges, where r is the number of terminals without a terminal ancestor. Thus the count is $G - 1$ only if $r = 1$. The LNG cover graph has at most $O(G^2)$ edges, and two-level subset families can attain a quadratic count. This is a worst-case distinction, not an assertion that every dataset has a dense LNG.

Proposition 1: group-level frontier coverage. Every eligible observed terminal is reachable from a terminal in $F(Q)$ by a path consisting entirely of eligible terminals in the prefix forest.

Proof. For an eligible terminal S , take the first eligible terminal A on its root-to- S trie path. If no earlier eligible terminal exists, $A \in F(Q)$ by definition. Consecutive terminals from A to S are joined by the nearest-terminal edges. Their label sets all contain A , hence contain Q . This also covers $A = S$ and $Q = \emptyset$. $\square$

Algorithm 1 Bitmap-backed prefix frontier (measured provider)

Require: query labels Q , label bitmaps I_a , proper terminal ancestors

```

1: if  $Q = \emptyset$  then
2:   return precomputed terminal roots of the prefix forest
3: if any query label has no posting then
4:   return  $\emptyset$ 
5: copy the smallest query-label posting into  $C_Q$ 
6: for each remaining query label  $a$  do
7:    $C_Q \leftarrow C_Q \cap I_a$ 
8:  $F \leftarrow \emptyset$ 
9: for each  $S \in C_Q$  do
10:  if no  $A \in \text{Anc}_p(S)$  belongs to  $C_Q$  then
11:    append the group identifier of  $S$  to  $F$ 
12: return  $F$ 

```

The proposition concerns the label plane. Lifting it to exhaustive vector reachability requires retaining every frontier group as a seed, each directed local graph being reachable from its seeded or incoming portals, and accessible materialized cross edges for each required group transition. Finite candidate capacity, approximate local graphs, pruning, and seed selection can violate those additional conditions. We therefore measure Recall; the proposition is not an ANN accuracy guarantee.

3.2 The Measured Entry Algorithm

The measured Trie provider stores, for each label a , a compressed bitmap I_a of groups containing a , and for each terminal S , its proper terminal ancestors $\text{Anc}_p(S)$. It intersects postings to obtain C_Q and tests only prefix ancestors to produce $F(Q)$ (Algorithm 1). This algorithm avoids comparing S against eligible terminals on unrelated branches. The bitmaps and ancestor lists are initialized before steady-state timing; per-query intersections and ancestor tests remain in the discovery time.

For comparison, both LNG-semantic providers first use a trie walk to collect a prefix-minimal candidate frontier, rather than

enumerate all eligible supersets. They then reduce this frontier under set containment. The original version sorts candidates by label cardinality and removes sets containing an already retained minimum. The optimized version uses cardinality buckets when the size span is small, with a sorting fallback; it still performs pairwise sorted-set containment checks for minimality. It preserves the original entry definition. The prefix provider changes that definition and the data structures used to compute it. Thus its timing advantage is not solely the effect of removing one sort.

Cost and output tradeoff. Let $c = |C_Q|$, $f = |F(Q)|$, and A_Q be the number of proper-ancestor membership probes actually performed. If $T_\cap$ is the cost of copying and intersecting compressed postings and t_ϵ bounds one bitmap membership probe, prefix discovery costs

$$O(|Q| + T_\cap + c + A_Q t_\epsilon + f). \quad (3)$$

For maximum label length h , $A_Q \leq ch$. Bitmap representation and compression affect $T_\cap$ and t_ϵ ; treating both as unit-time set operations would hide substantial work. The stored ancestor lists require up to $O(Gh)$ entries, in addition to postings and trie storage. The empty predicate uses the precomputed root frontier in $O(r)$ output time.

For the LNG providers, let c_{LNG} be the number of frontier candidates, T_{cand} their discovery time, and P_Q the number of pairwise minimality tests. With sorted label sets of length at most h , their cost is bounded by

$$O(T_{\text{cand}} + \text{Order}(c_{\text{LNG}}) + P_Q h), \quad P_Q \leq c_{\text{LNG}}(c_{\text{LNG}} - 1)/2, \quad (4)$$

where ordering is $O(c_{\text{LNG}} \log c_{\text{LNG}})$ for sorting or $O(c_{\text{LNG}} + \Delta)$ for cardinality buckets over span Δ . The prefix algorithm restricts which relations are tested, but may return more groups and hence increase seed distance work and graph traversal. We report these downstream costs separately.

4 PREDICATE-CERTIFIED BLOCK NAVIGATION

4.1 Two Granularities, One Vector Universe

Prefix-frontier entry changes how search starts; blocks change what it can search locally. Physical level 0 contains exact groups and their vector graphs. Physical level 1 adds vector graphs over groups aggregated into blocks. Within an authorized block, proximity edges can connect vectors from different exact groups directly, avoiding a requirement to follow the fine label relation for every transition. The base remains necessary for queries that cannot certify a useful coarse region.

We call this a 1L index: one overlay above the base, hence two physical levels. A 2L index has levels 0, 1, and 2. More generally, an m -overlay index has $m + 1$ physical levels. All levels reference the same vector identifiers, but their block counts, represented membership, and vector-edge counts can differ. This is not HNSW’s randomized subsampling of vector vertices, nor a bottom-up search from one level to the next.

A configuration is (m, T, τ, e, r) : overlay count, increasing thresholds, topology per physical level, entry provider, and routing policy. In particular $\tau_0 \in \{L, T\}$ is an independent choice; an LNG base is not part of the definition of ML-UNG. Historical

Table 1: Canonical topology notation; L means LNG and T means prefix Trie. Each upper level is an independently built block overlay.

Code	Level 0	Level 1	Level 2
$\emptyset L[T]$	Trie groups	absent	absent
$1L[T L]$	Trie groups	LNG blocks	absent
$2L[T LT]$	Trie groups	LNG blocks	Trie blocks

Algorithm 2 One independent block partition

Require: canonical trie, terminal weights $w(v)$, threshold T

```

1: for nodes  $v$  in postorder do
2:    $u(v) \leftarrow w(v) + \sum_{c \in \text{child}(v)} u(c)$ 
3:   if  $v \neq \text{root}$  and  $u(v) > T$  then
4:     mark  $v$  as a block root;  $u(v) \leftarrow 0$ 
5: assign each terminal to its nearest marked ancestor (including
   itself), if one exists
6: return roots and their direct-member lists  $P_b$ 
```

plot labels $1L$ -LNG and $2L$ -LT denote the fixed-LNG-base controls $1L[L|L]$ and $2L[L|LT]$. The full notation always lists the base before the separator.

4.2 Independent Mass Partitions

Each terminal has weight equal to its group size. For threshold T_ℓ , a bottom-up pass accumulates uncovered point mass. A non-root node becomes a block root when its accumulated mass exceeds T_ℓ ; that mass is then removed from its parent’s uncovered total. A block owns terminals below its root up to, but excluding, another block root at the same level.

Direct-member sets are disjoint within a layer, but may form only a partial cover: residual groups below no emitted root remain represented by the base. Threshold T_ℓ is an emission threshold, not a maximum block size. A large exact group or branching node can yield a block exceeding it. Separate partition passes for $T_1 < T_2 < \dots$ leave earlier levels unchanged; they do not imply nested direct-member sets or identical coverage at all levels.

4.3 Certification and Connectivity

Block b stores its root prefix R_b , direct members P_b , a vector entry point, and adjacency with an explicit owner level. It is authorized when $Q \subseteq R_b$. Since $R_b \subseteq A_i$ for every $i \in P_b$, the certificate implies $Q \subseteq A_i$. An unauthorized block may contain eligible points; inability to certify it says nothing about those points’ individual labels.

Within a block, the proximity graph need not respect exact-group boundaries. Between blocks at the same level, either a prefix relation or the LNG cover relation specifies legal transitions to more specific root-label sets. Only the layer’s direct-member points are used to realize its vector edges. Both relations preserve eligibility. Prefix edges are sparse, but are not necessarily a subset of the LNG cover edges: an intermediate observed set on another prefix branch can remove a cover edge that the prefix forest contains. Prefix reachability is a subset of containment reachability.

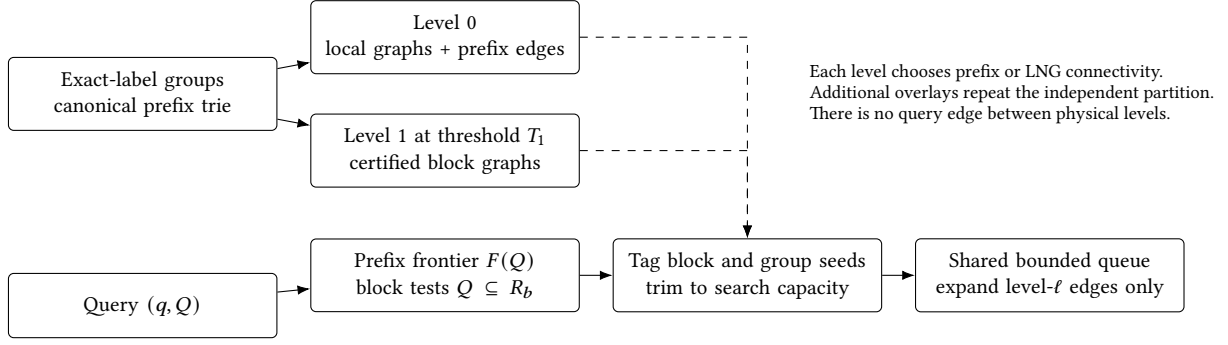


Figure 3: The simplest block instance has physical levels 0 and 1. Solid arrows show construction or query flow; dashed arrows supply adjacency and membership metadata. The optional router can invoke base search before overlay seeding. A candidate’s level is fixed at insertion and inherited by its successors.

Algorithm 3 Terminal-prefix topology at a fixed physical level

Require: canonical trie with the level’s groups or block roots marked

```

1: push (root, ⊥); E ← ∅
2: while stack not empty do
3:   pop (v, a)
4:   if v is marked then
5:     if a ≠ ⊥ then
6:       add (a, v) to E
7:     a ← v
8:   for each child c of v do
9:     push (c, a)
10: return E

```

This is the reason to test topology quality rather than infer it from edge counts. Algorithm 3 is used on groups at level 0 and on block roots at an overlay level.

5 QUERYING CERTIFIED BLOCKS

5.1 Seeding and Optional Routing

Entry discovery returns actual group identifiers. The prefix instance uses Algorithm 1; LNG-base controls use one of the two minimal-superset providers. These contracts are separate from block authorization and vector entry-point setup.

An ungated configuration uses every authorized block as a potential seed. The evaluated two-overlay DRH router instead requires at least one authorized level-2 block. DRH-v2 additionally tests its total direct-member mass against a derived threshold. If the test fails, the query calls the base-only search with the same base topology and entry provider. In the general implementation the gate uses the highest authorized physical level at least 2, not necessarily the highest materialized level. The two notions coincide in the evaluated two-overlay case.

For an accepted multilayer query, authorized blocks contribute their stored entry points tagged with their physical level. Each group entry receives the finest authorized overlay owning that group, or level 0 if no such overlay exists. The measured initializer prioritizes block seeds when trimming to capacity, then fills

Algorithm 4 Level-local containment search

Require: (q, Q, K) , base and overlays H , provider e , route r , capacity L_s

```

1: if H has no overlays then
2:   return BASESEARCH(q, Q, e, L_s)
3: B_Q ← {b : Q ⊆ R_b}
4: if r is gated then
5:   find highest authorized physical level h ≥ 2, if any
6:   if no such h, or ∑_{b∈B_Q: ℓ(b)=h} |P_b| < T_gate then
7:     return BASESEARCH(q, Q, e, L_s)
8: discover group entries with e and assign seed levels
9: create authorized block seeds and group seeds
10: deduplicate by (i, ℓ), mark seeds seen, and score them before trimming
11: prioritize block seeds; trim to L_s and initialize the queue
12: while an unexpanded state (i, δ, ℓ) remains do
13:   mark the closest unexpanded state expanded; retain it as a result candidate
14:   read only level-ℓ adjacency of i
15:   for each permitted edge to j at level ℓ do
16:     if (j, ℓ) is unseen then
17:       mark it seen; offer (j, dist(q, x_j), ℓ) to the queue
18: return the closest at most K distinct vector identifiers retained

```

remaining space with group seeds. It does not preserve a second base seed for every retagged entry. Therefore the existence of the base graph is not a coverage guarantee inside an accepted multilayer search.

5.2 Level-Local Search

Search states are (i, δ, ℓ) : vector identifier, query distance, and physical level. Visited state is indexed by (i, ℓ) , while the answer set is deduplicated by vector identifier. The shared queue has a configured active capacity L_s . Algorithm 4 describes the containment path with root-prefix authorization used in the reported experiments.

Proposition 2: predicate safety. Assume all group seeds are eligible and every overlay-tagged group seed is a direct member of an authorized owner block; base cross-group edges preserve strict containment; block seeds are direct members of authorized blocks; and each overlay edge is expanded only with an authorized source owner matching its state level, and stays within its direct-member block or follows a same-level strict-superset block transition. Then every vector whose query distance is computed in this search is eligible.

Proof. Eligible group seeds and certified block seeds satisfy Q . An intra-group edge preserves the label set. A base cross edge moves to a superset, preserving Q . Inside a certified block every direct member satisfies Q ; a cross-block edge moves to a root containing the source root and hence Q . Induction over generated states proves the claim. $\square$

The implementation relies on these construction and authorization invariants; it does not implement the guarantee by filtering every neighbor after computing its distance. The authoritative experiments enable root-prefix authorization, including the one-overlay case. Other predicate operators or legacy authorization modes require a separate proof.

Proposition 3: layer isolation and fallback. A successor inherits its source state’s physical level, and adjacency owned by other levels is rejected. Induction therefore gives level-local expansion with no promotion or fallback. When the router rejects an overlay before seeding, deterministic execution with identical base entries and budget reproduces the base-search path. This equivalence excludes the extra routing time, and does not assert equivalence after overlay admission.

These properties establish eligibility and ownership, not monotone Recall or latency. Additional seeds and retagging can displace useful states in a bounded queue. In particular, Proposition 1 cannot be applied after frontier seeds have been dropped without reinstating its coverage assumptions.

5.3 Query Cost

Let D_q be distance evaluations, E_q inspected adjacency entries, A_q candidate insertion attempts, and P_q state expansions. Let T_{entry} include group discovery and vector-seed setup, excluding distances already counted in $D_q d$. All scored seeds, including discarded ones, contribute to D_q . Direct certification costs $T_{\text{auth}} = O(\sum_b(|Q| + |R_b|))$ plus metadata traversal. The measured default array queue can scan its active capacity even when an insertion is rejected, so a safe bound is

$$T_q = O(T_{\text{entry}} + T_{\text{auth}} + D_q d + E_q + (A_q + P_q)L_s + L_s \log L_s). \quad (5)$$

This bound exposes why fewer group edges need not lower latency: the frontier, required budget, distances, and queue operations can all change. The optional lazy heap implementation has different costs; retained stale records can grow with successful insertions, so neither its storage nor its operation cost is correctly summarized as strictly $O(L_s)$ and $O(\log L_s)$. Per-level visited state, seed buffers, and queue backing storage contribute query workspace beyond the bounded active queue. A vector reached at different levels can incur repeated distance evaluations.

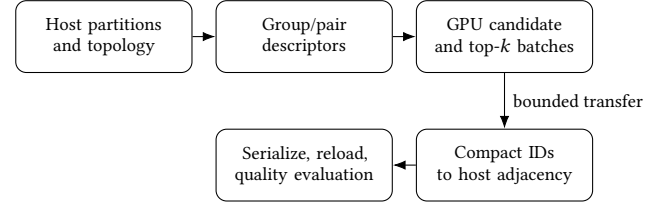


Figure 4: Construction dataflow. The GPU accelerates declared numerical stages; topology and output adjacency retain host ownership. Structural validation and query-quality evaluation are distinct checks.

6 CONSTRUCTION AND HIERARCHY CONFIGURATION

6.1 Batched Construction of Irregular Graphs

The base and overlays share two computational tasks: constructing a local vector graph for a variable-size member set, and selecting cross edges for an irregular pair of member sets. Launching a GPU operation for every tiny group repeats preparation and launch costs. Conversely, a few large blocks can dominate local graph construction. Our pipeline separates host metadata, memory-bounded GPU batches, and final host adjacency so that each cost remains visible.

Local graphs. The hybrid hierarchy builder handles small blocks with exact top- R graphs (or the configured bounded-complete alternative), medium blocks with sampled CPU Vamana, and large blocks with GPU neighbor-candidate refinement followed by diversity pruning and degree repair. The fixed thresholds are 2048 and 8192 points, with 256 sampled candidates for the middle route. The CUDA candidate path uses a FastGrnd-style adaptation of NN-Descent [3]; pruning follows relative-neighborhood principles [15]. These implementation settings are shared by compared structures. A separate full-GPU profile sends all blocks above the small completion threshold to CUDA. Neither route consults query Recall.

Cross edges. Each permitted group or block transition induces nearest-target tasks. Descriptors identify source and target member sets, dimensions, and degree budgets. A conventional GPU path computes SGEMM distance tiles and runs top- k separately; the fused path evaluates distances and selects target identifiers without materializing the complete score matrix in host memory. This uses the same distance/selection bottleneck exploited by GPU similarity search [7], specialized to many irregular tasks. Vectors and norms remain resident when they fit the configured device budget; otherwise bounded chunks are streamed. Only identifiers are transferred when downstream adjacency does not consume distances.

The base build uses the same task separation on exact groups. In the accelerated base profile, optimized host metadata, CUDA local graphs, fused cross edges, and a configured CPU additional-edge completion path jointly determine construction time. Consequently a full-builder speedup cannot be attributed to fusion alone.

Algorithm 5 Construction of one declared hierarchy plan

Require: vectors, label trie, plan (T_ℓ, τ_ℓ) , degree and memory budgets

- 1: **for** each level ℓ independently **do**
- 2: partition direct members and build the declared block topology
- 3: **for** each block b **do**
- 4: select its declared CPU/GPU local-graph route by member count
- 5: generate candidates, prune, and apply the degree-repair policy
- 6: form cross-edge task descriptors from the block topology
- 7: **for** each descriptor batch fitting device memory **do**
- 8: run the declared CPU or GPU distance/top- k backend
- 9: append compact identifiers to host adjacency with level owners
- 10: record effective backend, fallbacks, time, and allocated memory
- 11: serialize, reload, and validate membership and edge ownership
- 12: evaluate resulting index quality separately at the declared Recall targets

6.2 Costs and Quality Contract

The abstract uncovered-mass partition costs $O(m|\mathcal{T}|)$ for m overlays, plus emitted membership traversal. The actual builder additionally sorts labels and member identifiers and constructs bitmaps, ancestor caches, and dense ownership maps. Prefix topology has linear edge count; the current generic overlay LNG builder tests source, target, and possible intermediate sets, giving an $O(B_\ell^3 h)$ worst-case bound for B_ℓ roots of maximum label length h . This implementation bound is different from its quadratic maximum output size.

For an exact cross-edge batch with tasks (U, V) , distance arithmetic is

$$\Theta\left(d \sum_{(U,V)} |U| |V|\right). \quad (6)$$

The number of selected identifiers is at most $\sum_{(U,V)} |U| \min(C, |V|)$ for target budget C . Fusing selection reduces intermediate traffic and launch overhead; it does not remove the underlying distance work. Approximate local-graph construction depends on candidate counts and refinement rounds, so we report actual work and time rather than assign it an exact-search bound.

Loaded hierarchy storage includes direct memberships and vector adjacency, plus $O(m(N + G))$ dense point/group ownership maps. Shared vector coordinates need not be duplicated per overlay. Device residency or streaming must fit the declared memory budget, including candidates and output buffers. Query workspace also includes per-level visited state, seed buffers, and queue back-ing storage.

Numerical backend execution, successful serialization, and edge-ownership checks establish that an index was built as requested. They do not establish equal ANN quality. In particular, approximate candidate generation or TF32 arithmetic can change adjacency. A speed-quality claim therefore requires re-querying

Algorithm 6 DRH structure and optional mass gate

Require: vector count N , local degree R , cross degree C

- 1: $T \leftarrow \text{nearestPow2}(\sqrt{N})$; $\rho \leftarrow \max(2, \text{round}(R/C))$
- 2: **while** $N/T \geq C$ and depth cap is not reached **do**
- 3: append (T, LNG) if $N/T > R$, otherwise (T, Trie)
- 4: $T \leftarrow \rho T$
- 5: **return** overlay plan and next-scale gate $T_{\text{gate}} = T$

each accelerated index, including the fastest hybrid build. The construction table reports measured time and resources; the current paper does not infer a quality-preserving speedup from those columns alone.

6.3 A Query-Calibration-Free Instantiation

Independent overlays introduce depth and scale choices. DRH uses a simple structural proxy $T + N/T$, whose continuous minimum is $\sqrt{N}$, to choose the first threshold. It then derives a growth ratio from local degree R and cross degree C :

$$T_1 = 2^{\lceil \log_2 \sqrt{N} + 1/2 \rceil}, \quad \rho = \max(2, \text{round}(R/C)), \quad T_{\ell+1} = \rho T_\ell. \quad (7)$$

It materializes scales while $N/T_\ell \geq C$. Ignoring the implementation depth cap and integer-overflow guard, the derived overlay count is

$$m = \max\left(0, 1 + \left\lceil \log_\rho \frac{N}{CT_1} \right\rceil\right). \quad (8)$$

An overlay uses LNG when $N/T_\ell > R$ and prefix topology otherwise. The coarsest graph is thus sparsified once the estimated block count fits within one local neighborhood budget. Here N/T_ℓ is a proxy, not the measured block count; threshold partitioning leaves residuals and can emit oversized blocks. Neither the proxy objective nor the topology rule is a theorem about query runtime.

For Amazon, $N = 602,453$, $R = 64$, and $C = 4$ give the plan 1024:LNG and 16384:Trie. DRH-v1 enables the multilayer backend when an authorized physical level $h \geq 2$ exists (level 2 in the evaluated plans); DRH-v2 also requires sufficient direct mass at the highest such level, using the next unmaterialized threshold. The held-out policy experiment uses a conservative LNG base. It does not automatically choose between LNG and Trie bases, and must not be substituted for the matched-provider factorial winners. The policy reads no query frequencies, latency, or Recall, but its per-query authorization and mass gate naturally depend on Q .

6.4 Maintenance Scope

Inserting a label set changes a canonical trie path and may change nearby terminal-prefix relations. This avoids maintaining all minimal-superset relations across unrelated branches at the label-plane level. It does not make full-index updates free: label postings and ancestor caches change, mass thresholds may trigger repartitioning, and local/cross-block vector graphs require repairs. A deletion can split a block or invalidate portals. The evaluated implementation is a static builder; incremental publication, concurrent query visibility, and update amortization remain unmeasured design questions. We claim no dynamic-index result from this structural observation.

7 EXPERIMENTAL METHODOLOGY

7.1 Questions

We ask: (Q1) When does the prefix topology with its covering frontier improve over the matched LNG system? (Q2) Which effects come from block depth and controlled topology changes? (Q3) Which profile counters explain these effects? (Q4) What construction time and resources do GPU-assisted overlays require? (Q5) How well does a query-calibration-free plan transfer relative to a frozen manual candidate set?

7.2 Execution Environment

The experiment host uses two Intel Xeon Platinum 8360Y processors (72 physical cores, 144 logical CPUs) and an NVIDIA L20 with 46,068 MiB reported device memory. Queries use 100 CPU workers. The manuscript audit records CentOS 7 and driver 550.54.14 in a separate hardware snapshot; this describes the observed host, while original binary hashes, per-case backend settings, and GPU-idle evidence remain in the frozen campaign manifests. GPU acceleration in the construction study does not imply GPU query execution.

7.3 Datasets and Workloads

The development dataset is Amazon with 602,453 vectors of dimension 768 and 482,387 observed label groups. Each Amazon workload contains 1,000 queries, uses $K = 10$, and has exact containment ground truth. The measured mean selectivities are 0.499249%, 0.903055%, 5.038117%, 9.906615%, 30.027242%, 60.047491%, 80.023699%, 95.019781%, and 99.000600%. Genome, Reviews, and VariousImgs are held out for the automatic-policy study; their exact workload directories and fingerprints are recorded in the artifact manifest. The 99% Amazon workload contains 697 empty predicates and 303 singleton predicates $\{1\}$. We retain this mixture because both are legal containment queries, and validate the empty-predicate path separately before timing.

7.4 Methods and Fairness

The principal 0L comparison is LNG topology with optimized-LNG entry discovery versus Trie topology with Trie entry discovery. The topology factorial fixes $T_1 = 1024$, $T_2 = 16384$, and ungated routing while enumerating both base topologies and every LNG/Trie upper-layer assignment through 2L. Within either base, replacing one upper letter is a controlled topology comparison because its entry provider is unchanged. Across bases we instead report a matched-provider system comparison, not a pure edge-topology effect. The implementation supports three entry strategies (original LNG, optimized LNG, and Trie); the current tables do not constitute a complete controlled three-provider latency ablation. Routed controls are outside the factorial because routing changes which search algorithm executes. For held-out automatic-policy evaluation, however, the exact gate is fixed for DRH and every manual-alternative hierarchy; the ungated DRH variant is reported separately.

We call the 0L LNG configuration *Plain* or *0L-LNG*. It preserves UNG’s exact-label groups, LNG cover relation, and unified vector graph, but it is not the unmodified upstream release: it uses our

optimized, semantically equivalent minimal-superset provider and the common refactored execution path. Within a fixed LNG base and provider, this is the control for adding overlays. Changing the base and its matched provider changes two mechanisms jointly. These measurements are not direct timings against the original UNG binary.

Within each comparison, query methods use the same data, query vectors, labels, exact ground truth, K , thread count, immutable search binary, graph backend, and Recall protocol. All primary query values use 100 query threads over the full frozen batch. The deadline evidence is explicitly screen-level: one cold run followed by two warm repeats, with QPS from warm-median batch time. Every repeat executes the same frozen query file in the same order; repeats are repeated measurements, not new random samples. We discard the cold repeat, derive primary QPS from the median warm batch time, and retain per-repeat values for dispersion estimates. Recall is not averaged to relax the quality target: every warm repeat at the selected operating point must meet the threshold. We select the smallest *measured* L_{search} for which every warm repeat reaches $\text{Recall}@10 \geq 0.90$; there is no interpolation or extrapolation. The screen pass locates crossings and a separate profile pass collects expensive counters. The current deadline artifact uses one discarded cold repeat and two warm repeats; a later formal pass is reserved for confidence intervals. If a method has no screen crossing, its crossing pass extends to the largest L_{search} measured for any method on that workload (or an explicitly declared common cap); only failure at that shared budget is reported as no crossing.

7.5 Metrics

Table 4 defines every quantity used in the query tables. Batch size B is workload-specific: 1,000 on Amazon and 2,864 or 3,000 on the held-out workloads (Table 2). Primary query metrics are warm-median batch latency and QPS. Bootstrap 95% confidence intervals are reserved for the later formal pass and are not claimed by the current deadline artifact.

Build timing uses process wall time through files on disk, followed by structural acceptance checks; kernel time is never substituted for builder wall time. The current construction evidence contains two measured repeats per hierarchy backend, after a cold run, and a separate memory-resource pass. The composed total adds independently measured base and sidecar stage medians; it is not one jointly timed multilevel build. Bootstrap intervals resample these small observed stage cohorts independently. They describe this exploratory sample and do not establish population-level uncertainty. A longer interleaved timing study remains necessary for a robust interval. GPU cases retain three consecutive idle-device snapshots before timing on this host, where an exclusive GPU lock is unavailable; the evidence establishes idle preflight, not lock-level exclusivity. Backend counters and reload checks validate execution and structure. The current timing table does not demonstrate matched-Recall quality for every accelerated index, especially the fastest hybrid configuration.

Table 2: Held-out datasets and query-independent DRH plans. Selectivity characterizes the frozen queries and is not an input to DRH.

Dataset	N	d	Workload	Queries	Selectivity	DRH plan
Genome	108,077	512	query_minlen5_cov0.1k	3,000	3.365%	256:LNG, 4096:TRIE
Genome	108,077	512	query_minlen2_cov1k	3,000	6.292%	256:LNG, 4096:TRIE
Reviews	288,065	384	query_minlen5_cov0.1k	3,000	0.200%	512:LNG, 8192:TRIE
Reviews	288,065	384	query_minlen2_cov1k	2,864	4.115%	512:LNG, 8192:TRIE
VariousImg	758,935	512	query_minlen2_cov5k	3,000	10.252%	1024:LNG, 16384:TRIE

Table 3: Measured Amazon crossing and screen caps.

Selectivity	Plain crossing L	Screen cap L
0.499%	2,000	2,000
0.903%	2,200	2,200
5.038%	3,200	5,000
9.907%	26,000	64,000
30.027%	120,000	150,000
60.047%	260,000	320,000
80.024%	320,000	602,453
95.020%	320,000	602,453
99.001%	400,000	602,453

7.6 Frozen Construction Profiles

8 RESULTS

8.1 Entry and Topology Choices

We first compare every base and overlay topology using matched entry providers. These 126 cells are measured Recall crossings or explicit no-crossing outcomes, not interpolated optima. They expose the operating regime of the prefix-entry design before the automatic policy is evaluated.

Complete base-by-overlay factorial. The workload-wise oracle selects 2L[T|LT] at 0.5%, 1%, 5%, and 95%; 0L[L] at 10% and 30%; 1L[T|L] at 60% and 80%; and 2L[T|TT] at 99%. Thus no static topology dominates every workload. Among configurations crossing the Recall target at all nine selectivities, 2L[T|LT] ranks first by geometric mean of QPS normalized to the per-workload oracle (0.868), but its worst fraction of the oracle is 0.482, at 30%. The matched-provider base-only comparison is equally non-monotone: 0L[T] is 6.03 $\times$, 2.77 $\times$, and 18.43 $\times$ faster at 0.5%, 1%, and 5%, respectively, but is 0.60 $\times$ at 10%, does not cross at 30% ($R_{\max} = 0.8882$), and is 0.62 $\times$ –0.33 $\times$ at 60%–95% (0.56 $\times$ at 99%). These are end-to-end system ratios because the two bases use different compatible entry providers.

The controlled upper-level comparisons reveal a more stable role for Trie. With a Trie base and LNG at level 1, changing only level 2 from LNG to Trie (2L[T|LL] $\rightarrow$ 2L[T|LT]) improves QPS in eight of nine workloads, by up to 1.251 $\times$; the remaining 80% point is effectively tied at 0.998 $\times$. In contrast, changing only the finer level 1 from LNG to Trie with level 2 fixed to LNG wins in five of nine workloads but loses 40.7%–52.9% at 10%, 60%, 80%, and 95%.

The evidence therefore favors Trie as a sparse coarse overlay more consistently than as the fine navigation topology, while retaining an LNG fallback for the 10%–30% region.

8.2 Depth and Fixed-Base Controls

The following controls keep the LNG base and optimized-LNG entry provider fixed when varying overlays. Historical labels abbreviate the base as defined in Table 1. Gated and ungated routing are separate configurations; a depth benefit under one routing policy cannot be transferred to another by renaming the index. The selected stage profile in Figure 2 shows discovery savings and their possible displacement into seed and graph work. Full profiles and fixed-base topology controls appear in the appendix; the one declared profile timeout is preserved.

8.3 Automatic Configuration and Transfer

Manual alternatives are a frozen finite candidate set. Per-workload winners are hindsight comparisons, not executable query-independent policies.

8.4 Construction

The completed held-out sidecars below are CPU query-enablement builds, not evidence of GPU hierarchy speedup. The supervisor process-envelope timing, which includes orchestration and checks, measured 190.61s for original CPU and 53.18s for accelerated_gpu, a 3.58 $\times$ process-envelope speedup. The composed table below instead uses the component builder-time summaries; the two timing boundaries must not be mixed. The campaign manifest contains 41 completed cases; the repeated hierarchy and composed results follow.

8.5 Automatic Versus Manual Hierarchies

The bounded held-out study compares DRH-v1 with five pre-registered manual alternatives. DRH-v2 was evaluated later as a separate same-binary routing ablation and is not substituted into this frozen candidate set. The manual comparison is restricted to this finite candidate set.

8.6 Structural Router Ablation

DRH-v2 is a same-binary ablation: it rejects an upper layer when direct mass at the highest authorized layer is below the next derived scale. The next-scale gate restores near-plain behavior on

Table 4: Measurement definitions. A “warm median” is taken across the two warm repeats in the current screen-level artifact; it is not a confidence interval. Per-query stage times and work counters are first averaged over all queries in one repeat and then summarized across warm repeats.

Reported quantity	Definition
Mean selectivity	$B^{-1} \sum_{j=1}^B \mathcal{E}(Q_j) /N$ for the frozen query batch; it describes the workload and is not an input to DRH.
Recall@10	For one repeat, the mean over queries of $ R_{10}(q_j, Q_j) \cap R_{10}^*(q_j, Q_j) /10$. A crossing requires every warm repeat at that executed point to reach 0.90.
L_{search}	Capacity of the bounded candidate queue. An equal-Recall operating point is the smallest executed L_{search} that crosses the Recall target; values between executed points are not interpolated.
Batch latency / QPS	Wall time to complete all B queries with 100 query threads, after the cold repeat is discarded; $QPS = 1000B/\text{batch latency (ms)}$.
A/B speedup	QPS_A/QPS_B , using each method’s selected equal-Recall crossing. A ratio above one favors A .
$NC / R_{\max}$	NC means no Recall crossing within the workload-wide measured L_{search} budget. $R_{\max}$ is the largest minimum-over-warm-repeats batch Recall observed in that budget.
ELS / entry / authorization	Mean milliseconds per query for Entry Label Set (entry-group) discovery, vector entry-point setup, and exact block-authorization plus routing work, respectively.
Graph / residual	Mean milliseconds per query inside bounded graph traversal, and the remaining measured query time outside the named stages.
Visited / edges / distances	Mean per-query counts of popped graph states, scanned adjacency entries, and vector-distance evaluations. Base and overlay edges are counted separately where enabled.

Table 5: Frozen base-graph construction profiles. “Optimized metadata” denotes bucketed minimal-superset discovery, optimized LNG construction, epoch-based descendants, and direct descendant coverage.

Profile	Intra-group graph	Group metadata	Vector cross-group edges
original_cpu	CPU Vamana	Original sort/LNG, hash-BFS descendants, topological coverage	Original CPU builder
current_cpu	CPU Vamana	Optimized metadata	CPU Vamana cross-edge builder
naive_gpu	CPU Vamana	Optimized metadata	Batched GPU SGEMM plus separate top- k
paper_fused	CPU Vamana	Optimized metadata	Descriptor-grouped fused distance/top- k with identifier-only writeback
accelerated_gpu	FastGrnd CUDA path	Optimized metadata	Fused GPU cross edges plus CPU exact-scan additional edges

Table 6: Frozen hierarchy-sidecar construction profiles. CPU completion is retained below the declared small-block threshold in every profile.

Profile	Intra-block route	Inter-block route
cpu	Small exact/complete, medium sampled Vamana, large CPU Vamana	CPU
hybrid_gpu_intra	Small exact/complete, medium sampled Vamana, large FastGrnd-style CUDA	CPU
hybrid_gpu_intra_inter	Same hybrid intra route	Fused CUDA top- k
full_gpu	Bounded completion, then CUDA	Fused CUDA-core top- k
full_gpu_wmma	Bounded completion, then CUDA	Fused TF32-WMMA top- k

Genome and the 0.2% Reviews workload, and improves VariousImg over DRH-v1, but it also rejects the useful overlay at 4.115% Reviews. Thus it reduces catastrophic overhead without establishing monotone dominance.

8.7 Mechanism Breakdown

The separate detailed-profile binary is explanatory only and does not contribute primary QPS. Edge counts are measured here; disabled light-stat counters are never interpreted as zero.

9 DISCUSSION AND RELATED WORK

9.1 What the Evidence Establishes

The full factorial answers which measured configuration wins on each Amazon workload; it does not supply a query-independent

oracle choosing those winners. Changing one upper topology at fixed base and entry provider is controlled. The 0L matched-provider comparison instead measures an integrated topology and entry-design choice. Its favorable low-selectivity results do not extend to all broad predicates. Fewer logical edges can reduce navigation alternatives, while a larger prefix frontier increases seed work. Additional overlays introduce authorization and seed competition even with isolated adjacency.

The automatic policy provides a reproducible starting configuration without query-performance calibration. It does not automatically select the base, and its mass gate is a structural heuristic rather than a latency model. All four evaluated datasets derive two overlays, so threshold/topology transfer is tested but depth-change performance is not. The VariousImg regression is retained.

Table 7: Best measured configuration at each Amazon selectivity. Each depth column is optimized only within that depth; the overall column ranges over all 14 configurations. Values are QPS at Recall@10 $\geq$ 0.90.

Sel.	Overall		Best 0L		Best 1L		Best 2L	
	Config.	QPS	Config.	QPS	Config.	QPS	Config.	QPS
0.5%	2L[T LT]	21707.89	0L[T]	14921.90	1L[T T]	17937.69	2L[T LT]	21707.89
0.9%	2L[T LT]	14278.50	0L[T]	11591.42	1L[T L]	12447.15	2L[T LT]	14278.50
5.0%	2L[T LT]	11718.35	0L[T]	10044.23	1L[T T]	10937.27	2L[T LT]	11718.35
9.9%	0L[L]	450.17	0L[L]	450.17	1L[T L]	289.73	2L[T LT]	275.64
30.0%	0L[L]	52.77	0L[L]	52.77	1L[T L]	27.66	2L[T TT]	27.14
60.0%	1L[T L]	1613.68	0L[L]	4.07	1L[T L]	1613.68	2L[T LT]	1598.42
80.0%	1L[T L]	648.35	0L[L]	1.80	1L[T L]	648.35	2L[T LL]	647.95
95.0%	2L[T LT]	564.63	0L[L]	1.50	1L[T L]	520.35	2L[T LT]	564.63
99.0%	2L[T TT]	267.67	0L[L]	0.79	1L[T T]	264.44	2L[T TT]	267.67

Table 8: Amazon equal-Recall comparison. Plain QPS is the raw throughput of 0L-LNG; every other numeric column is QPS(method)/QPS(Plain) at that method’s smallest measured point whose two warm repeats both reach Recall@10 $\geq$ 0.90. NC means no crossing in the shared measured budget.

Selectivity	Plain QPS	0L-Trie/plain	1L-LNG/plain	2L-LT ungated/plain	2L-LT DRH-v1/plain
0.499%	2476.35	6.03×	NC	NC	0.86×
0.903%	4191.04	2.77×	NC	NC	0.99×
5.038%	544.93	18.43×	NC	NC	0.85×
9.907%	450.17	0.60×	NC	NC	0.64×
30.027%	52.77	NC	NC	NC	NC
60.047%	4.07	0.62×	13.89×	15.09×	14.40×
80.024%	1.80	0.33×	20.33×	27.57×	24.22×
95.020%	1.50	0.33×	27.99×	26.46×	26.70×
99.001%	0.79	0.56×	19.57×	21.34×	18.78×

Table 9: Automatic DRH CPU sidecar construction.

Dataset	Wall s	Blocks	Upper	Edges
Genome	44.2	264	18	10,088,461
Reviews	120.4	217	16	27,590,479
VariousImg	633.2	190	10	80,034,774

A submission-strength evaluation still needs current-protocol external baselines, longer repeated timing, and query-quality checks for each accelerated construction profile. Historical external runs use a different protocol and are not merged with the factorial. Predicate safety here concerns static AND-containment indexes with root-prefix authorization; extending the predicate language or update model requires a new contract.

9.2 Filtered and Range-Constrained Graph Search

HNSW [10] and DiskANN [5] organize vector proximity for bounded ANN search. Filtered-DiskANN [4] incorporates labels

during graph construction. ACORN [11] supports predicate-agnostic graph search with expanded neighborhoods and compressed construction; its design includes a configurable low-selectivity pre-filtering fallback. Its operating regime therefore depends on both the graph and the fallback configuration.

UNG [2] is the immediate structural baseline. Its exact-label groups, minimal-superset LNG, and vector realizations jointly preserve containment-directed navigation. We expose a different entry/topology contract: a terminal-prefix forest with its covering frontier. The forest has fewer permitted relations than general containment, but its edges are not necessarily an edge-subset of the LNG cover graph. We then aggregate compatible groups into independently authorized blocks. HNSW instead samples vector vertices across geometric layers; our scales change the label-space navigation unit.

SeRF [24], iRangeGraph [19], and UNIFY [9] compress, compose, or combine interval-specialized graphs. Dynamic range systems extend such structures to arrival and update workloads [13, 22]. Their ordered scalar intervals inspire explicit granularity, but arbitrary label containment forms a partial order and requires a different authorization argument.

Table 10: Repeated Amazon hierarchy-sidecar construction measurements. Resource values come from a separate profiling run and do not enter timing medians.

Backend	Repeats	Wall median (s)	CV	Speedup/CPU	Peak RSS (MiB)	Peak GPU (MiB)
cpu	2	1000.64	0.005	1.00×	5001.5	0.0
full_gpu	2	105.90	0.007	9.45×	4178.1	2286.0
full_gpu_wmma	2	1011.13	0.000	0.99×	4223.5	2362.0
hybrid_gpu_intra	2	90.49	0.008	11.06×	4880.7	1598.0
hybrid_gpu_intra_inter	2	132.94	0.006	7.53×	4255.6	2286.0

Table 11: Composed end-to-end construction: accelerated base stage plus hierarchy sidecar versus the original CPU base builder. Stage medians are measured independently; timings do not establish matched-Recall quality.

Hierarchy backend	Stage repeats	Original CPU (s)	Composed (s)	Speedup	Exploratory 95% CI
cpu	2	190.38	1051.42	0.18×	[0.18, 0.18]
full_gpu	2	190.38	156.69	1.21×	[1.20, 1.23]
full_gpu_wmma	2	190.38	1061.91	0.18×	[0.18, 0.18]
hybrid_gpu_intra	2	190.38	141.27	1.35×	[1.33, 1.36]
hybrid_gpu_intra_inter	2	190.38	183.72	1.04×	[1.03, 1.05]

Table 12: Screen-level 2L-LT/DRH-v1 results at Recall@10 $\geq$ 0.90.

Dataset	Selectivity	Plain QPS	DRH-v1/plain	DRH-v1/best manual	Best manual plan
Genome	3.365%	153.65	0.954×	1.008	256,4096:trie,trie
Genome	6.292%	85.45	0.982×	1.007	256:lng
Reviews	0.200%	19319.56	1.014×	0.997	1024,16384:lng,trie
Reviews	4.115%	354.78	1.004×	0.980	1024,16384:lng,trie
VariousImg	10.252%	2657.04	0.205×	0.218	1024:lng

Table 13: One fixed DRH-v1 plan versus one fixed manual plan per held-out dataset (geometric-mean QPS).

Dataset	Workloads	DRH-v1 QPS	DRH-v1/manual
Genome	2	110.86	1.013
Reviews	2	2642.45	0.989
VariousImg	1	543.74	0.218

Curator [6] shares label-specialized virtual indexes through a clustering tree. Its temporary indexes for conjunctions can be assembled by qualified-ID ordering and partition lookup without distance computation; its reported two-label AND results are close to pre-indexed predicates. Our blocks instead have fixed canonical prefix membership and use a root-prefix certificate to authorize traversal. FAVOR [14] selects a filter-first brute-force path below an estimated selectivity threshold and an exclusion-distance HNSW path otherwise. Its empirically chosen threshold and reported dataset-dependent crossover motivate studying multiple

regimes; the absolute candidate mass remains relevant to the cost of its scan route.

Elastic Index Selection [20] and SIEVE [8] choose and share collections of indexes. Their selection problem differs from choosing block scales inside one containment-oriented system. Recent LSSG [17], an arXiv preprint, stratifies edges by label-set similarity and traverses multiple tiers. Thus merely adding label-aware layers is not our novelty: our distinctive mechanism is direct-member block aggregation, exact root-prefix authorization, and states restricted to one physical adjacency level. Query-aware Routing [18], also a preprint, uses an offline performance table and a learned Recall model; DRH deliberately uses neither, accepting the resulting loss of adaptability. These preprints are distinguished from accepted SIGMOD/PVLDB publications.

9.3 Construction and Maintenance

NN-Descend [3] and relative-neighborhood pruning [15] provide building blocks for proximity graphs. Database work improves construction with tree/graph hybrids, LSH-assisted insertion, monotonic neighborhoods, and revised pruning [1, 12, 21, 23].

Table 14: Same-binary DRH-v2 screen results. Plain is measured in this routing cohort; it need not match the earlier DRH-v1 cohort.

Dataset	Selectivity	Plain QPS	DRH-v1 QPS	DRH-v2 QPS	v2/plain
Genome	3.365%	143.73	144.50	142.57	0.992×
Genome	6.292%	82.74	83.73	83.16	1.005×
Reviews	0.200%	18210.79	17445.16	18335.39	1.007×
Reviews	4.115%	361.63	368.07	351.49	0.972×
VariousImg	10.252%	2633.06	547.01	884.18	0.336×

Table 15: Detailed profile at each held-out dataset’s broader workload (ms/query and mean work per query). Active is the fraction of all queries that execute the multilayer backend, summarized over warm repeats.

Dataset	Method	Graph ms	Visited	Edges	Distances	Active
Genome	0L-LNG (Plain)	0.124	123.8	154.3	4538.9	0.0%
Genome	2L-LT/DRH-v1	0.199	185.1	591.8	4601.4	1.8%
Reviews	0L-LNG (Plain)	6.756	4415.8	15126.2	8239.3	0.0%
Reviews	2L-LT/DRH-v1	6.504	3500.1	25605.4	7312.0	18.2%
VariousImg	0L-LNG (Plain)	35.098	18656.0	119045.5	18673.1	0.0%
VariousImg	2L-LT/DRH-v1	172.065	27356.1	251144.0	27393.7	37.4%

Starling [16] co-designs navigation and storage for disk I/O. GPU similarity search [7] emphasizes distance and selection throughput. Our contribution is to batch the irregular local-graph and cross-edge tasks induced by a metadata hierarchy, retaining an explicit host-output boundary. The static experiments do not establish dynamic-update cost or numerical-quality equivalence across construction backends.

10 CONCLUSION

Entry discovery and graph topology form a joint contract in label-filtered ANN. A prefix frontier covers all eligible terminal branches at the label plane, while independently partitioned blocks admit coarser navigation when a root-prefix certificate holds. Level-tagged states make the adjacency used by each expansion explicit. These structural properties explain the design; measured Recall and timing establish where it is useful. The Amazon factorial shows both substantial improvements and topology-dependent regressions. DRH supplies a query-calibration-free starting plan, and batched GPU construction reduces measured build cost. Neither is claimed to dominate manual tuning or preserve quality without its corresponding evidence.

Evidence provenance. Full paths and SHA256 digests for 27 validated inputs are stored in the deadline artifact manifest; their ordered aggregate digest begins 7e05bfbd4d08b7b.

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A COMPLETE TOPOLOGY AND MEASUREMENT TABLES

Table 16: Zero- and one-overlay configurations in the complete Amazon base-by-overlay topology screen at Recall@10 $\geq$ 0.90. A code $mL[B|U]$ lists base topology B before the separator and upper topologies U from fine to coarse. L is LNG and T is Trie. The base uses its matched entry provider: optimized-LNG for $B = L$ and Trie for $B = T$. NC(r) gives maximum minimum-warm-repeat Recall when there is no crossing in the shared measured budget.

Sel.	0L[L]	0L[T]	1L[L L]	1L[L T]	1L[T L]	1L[T T]
0.499%	2476.35	14921.90	NC (0.888)	NC (0.888)	15177.24	17937.69
0.903%	4191.04	11591.42	NC (0.857)	NC (0.846)	12447.15	12144.74
5.038%	544.93	10044.23	NC (0.893)	NC (0.883)	9105.22	10937.27
9.907%	450.17	270.63	NC (0.874)	NC (0.848)	289.73	146.19
30.027%	52.77	NC (0.888)	NC (0.836)	NC (0.828)	27.66	10.50
60.047%	4.07	2.51	56.52	56.70	1613.68	839.32
80.024%	1.80	0.59	36.65	37.08	648.35	285.42
95.020%	1.50	0.49	42.06	36.82	520.35	265.13
99.001%	0.79	0.44	15.40	14.34	253.23	264.44

Table 17: Two-overlay configurations in the complete Amazon base-by-overlay topology screen at Recall@10 $\geq$ 0.90. A code $mL[B|U]$ lists base topology B before the separator and upper topologies U from fine to coarse. L is LNG and T is Trie. The base uses its matched entry provider: optimized-LNG for $B = L$ and Trie for $B = T$. NC(r) gives maximum minimum-warm-repeat Recall when there is no crossing in the shared measured budget.

Sel.	2L[L LL]	2L[L LT]	2L[L TL]	2L[L TT]	2L[T LL]	2L[T LT]	2L[T TL]	2L[T TT]
0.499%	NC (0.888)	NC (0.888)	NC (0.888)	NC (0.888)	19158.52	21707.89	20593.10	17565.42
0.903%	NC (0.857)	NC (0.857)	NC (0.846)	NC (0.846)	13242.16	14278.50	14062.30	12353.41
5.038%	NC (0.893)	NC (0.893)	NC (0.884)	NC (0.884)	9569.96	11718.35	11082.17	10046.26
9.907%	NC (0.848)	NC (0.849)	NC (0.822)	NC (0.823)	270.57	275.64	135.16	139.26
30.027%	NC (0.839)	NC (0.839)	NC (0.833)	NC (0.832)	24.92	25.42	27.05	27.14
60.047%	56.37	61.40	57.00	56.42	1565.04	1598.42	849.99	844.65
80.024%	35.44	49.70	35.88	35.94	647.95	646.83	305.37	313.61
95.020%	39.61	39.75	36.99	36.68	451.32	564.63	267.67	269.48
99.001%	14.69	16.79	14.41	14.28	253.95	257.77	265.85	267.67

A.1 Amazon Stage and Work Breakdown

This profile-only pass reuses each method’s measured performance operating point. Rows marked max use the best measured point because Recall@10 did not cross 0.90; they explain work but are not equal-Recall speed comparisons. Timeout denotes a declared profile timeout at the fixed 3,300-second per-case cap; no partial counters are used.

Table 18: 1L overlay-topology ablation at fixed $T_1 = 1024$, level-0 LNG, and optimized-LNG entry. QPS is reported only at a measured Recall@10 ≥ 0.90 crossing; $R_{\max}$ is the largest minimum warm-repeat Recall in the shared measured budget.

Selectivity	1L-LNG QPS	1L-Trie QPS	Trie/LNG	LNG $R_{\max}$	Trie $R_{\max}$
0.499%	NC	NC	–	0.8878	0.8878
0.903%	NC	NC	–	0.8567	0.8464
5.038%	NC	NC	–	0.8928	0.8835
9.907%	NC	NC	–	0.8742	0.8477
30.027%	NC	NC	–	0.8359	0.8283
60.047%	56.52	56.70	1.003×	0.9511	0.9503
80.024%	36.65	37.08	1.012×	0.9346	0.9309
95.020%	42.06	36.82	0.875×	0.9289	0.9259
99.001%	15.40	14.34	0.931×	0.9325	0.9301

Table 19: 2L topology ablation at fixed $T_1 = 1024$, $T_2 = 16384$, level-0 LNG, optimized-LNG entry, and ungated routing. In 2L-XY, X is the level-1 topology and Y the level-2 topology. QPS appears only at a measured Recall@10 ≥ 0.90 crossing.

Sel.	2L-LL QPS	2L-LT QPS	2L-TL QPS	2L-TT QPS	LL $R_{\max}$	LT $R_{\max}$	TL $R_{\max}$	TT $R_{\max}$
0.499%	NC	NC	NC	NC	0.8878	0.8878	0.8878	0.8878
0.903%	NC	NC	NC	NC	0.8567	0.8567	0.8464	0.8464
5.038%	NC	NC	NC	NC	0.8929	0.8928	0.8836	0.8836
9.907%	NC	NC	NC	NC	0.8484	0.8492	0.8220	0.8229
30.027%	NC	NC	NC	NC	0.8385	0.8393	0.8333	0.8324
60.047%	56.37	61.40	57.00	56.42	0.9529	0.9538	0.9523	0.9525
80.024%	35.44	49.70	35.88	35.94	0.9357	0.9362	0.9341	0.9313
95.020%	39.61	39.75	36.99	36.68	0.9306	0.9310	0.9294	0.9271
99.001%	14.69	16.79	14.41	14.28	0.9338	0.9346	0.9325	0.9313

Table 20: Amazon detailed profile. Times are ms/query; work counters are means/query.

Sel.	Op.	Method	ELS	Entry	Auth.	Graph	Visited	Edges	Distances
0.499%	cross	0L-LNG	20.352	0.195	0.000	2.076	1966.8	23458.9	2150.2
0.499%	cross	0L-Trie	0.645	1.769	0.000	0.868	988.6	11047.0	2764.4
0.499%	max	1L-LNG	19.665	0.216	0.015	4.418	1860.2	23296.2	2043.9
0.499%	max	1L-Trie	16.188	0.224	0.012	4.578	1859.2	23263.7	2042.9
0.499%	cross	2L-LT/DRH-v1	20.455	0.194	0.003	2.109	1968.1	23464.7	2151.5
9.907%	cross	0L-LNG	36.631	0.646	0.000	98.747	25024.2	189382.5	25402.9
9.907%	cross	0L-Trie	3.778	14.744	0.000	323.081	38926.2	351908.0	44682.9
9.907%	max	1L-LNG	33.451	0.645	0.017	644.477	34782.5	821915.0	35184.9
9.907%	max	1L-Trie	32.969	0.576	0.015	587.339	32344.1	757831.0	32746.5
9.907%	cross	2L-LT/DRH-v1	31.648	0.685	0.023	301.608	28532.4	449218.0	28936.2
30.027%	cross	0L-LNG	251.380	3.963	0.000	1376.910	91953.6	580600.5	93654.0
30.027%	max	0L-Trie	8.381	30.629	0.000	2420.260	88466.5	497828.0	98769.0
30.027%	max	1L-LNG	220.485	1.887	0.017	1765.145	61473.4	1112780.0	63220.3
30.027%	max	1L-Trie	226.654	1.963	0.015	1662.880	58664.4	1021480.0	60411.4
30.027%	max	2L-LT/DRH-v1	234.946	2.091	0.013	1755.115	64374.7	978653.5	66124.7
60.047%	cross	0L-LNG	2123.720	33.187	0.000	21357.550	254657.0	1988540.0	265484.5
60.047%	cross	0L-Trie	10.670	63.353	0.000	39255.750	261406.5	1453445.0	278174.0
60.047%	cross	1L-LNG	1415.925	8.882	0.017	200.452	45762.3	218706.0	56600.6
60.047%	cross	1L-Trie	1465.900	8.881	0.013	192.047	44045.4	200138.0	54883.6
60.047%	cross	2L-LT/DRH-v1	1408.655	9.064	0.017	195.087	45607.8	215753.5	56454.1
95.020%	cross	0L-LNG	4918.875	61.074	0.000	89240.100	451225.0	3967260.0	467715.0
95.020%	timeout	0L-Trie	—	—	—	—	—	—	—
95.020%	cross	1L-LNG	2307.095	12.576	0.018	113.150	44157.5	170659.0	60663.6
95.020%	cross	1L-Trie	2313.385	13.444	0.014	299.972	67643.4	309628.0	84149.5
95.020%	cross	2L-LT/DRH-v1	2317.090	12.571	0.024	113.679	43995.0	170239.0	60513.8

B COMPLETE RECALL-THROUGHPUT CURVES

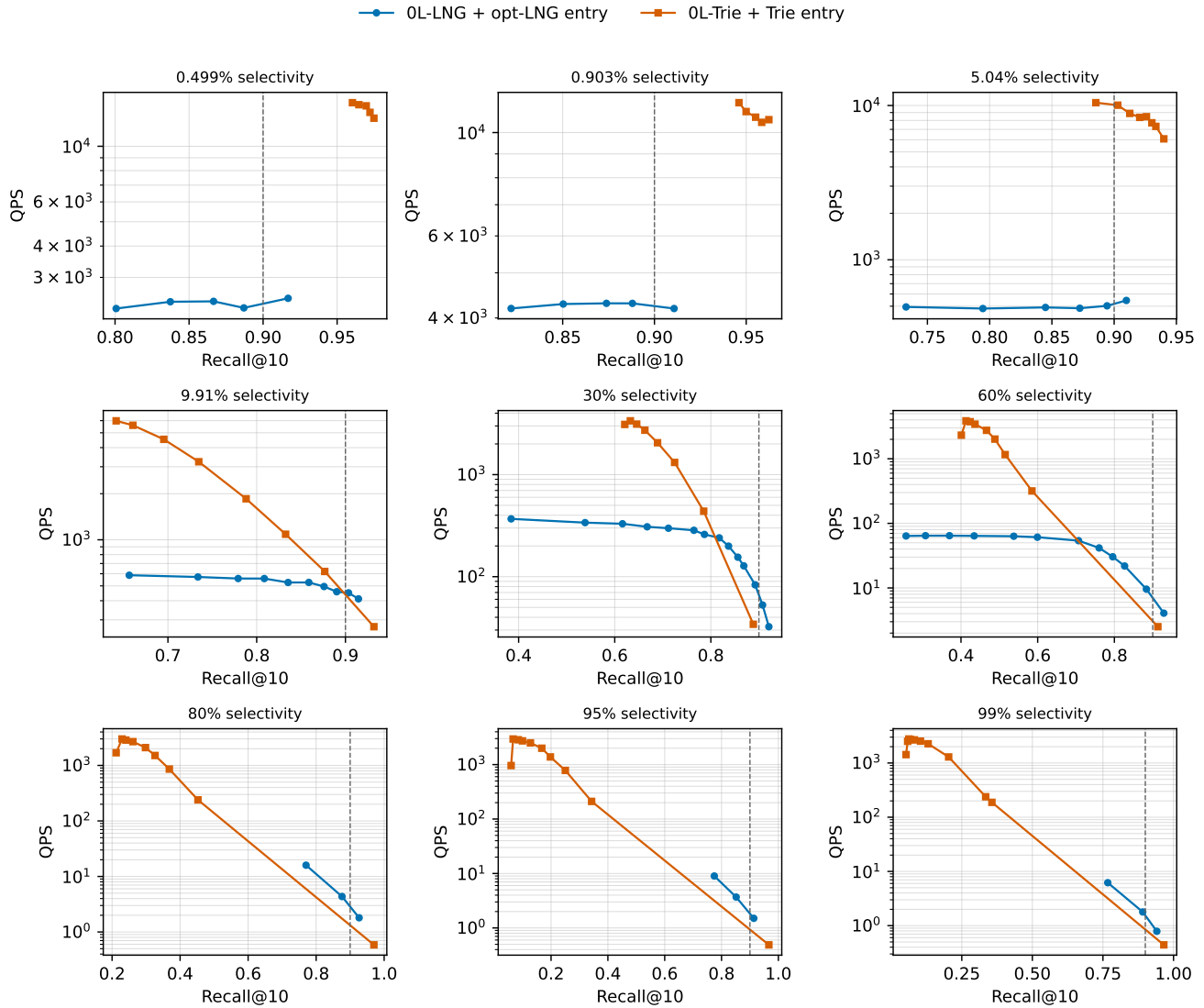


Figure 5: Measured Recall@10-QPS curves for the 0L-LNG/0L-Trie comparison. Markers are executed points; curves are not interpolated.

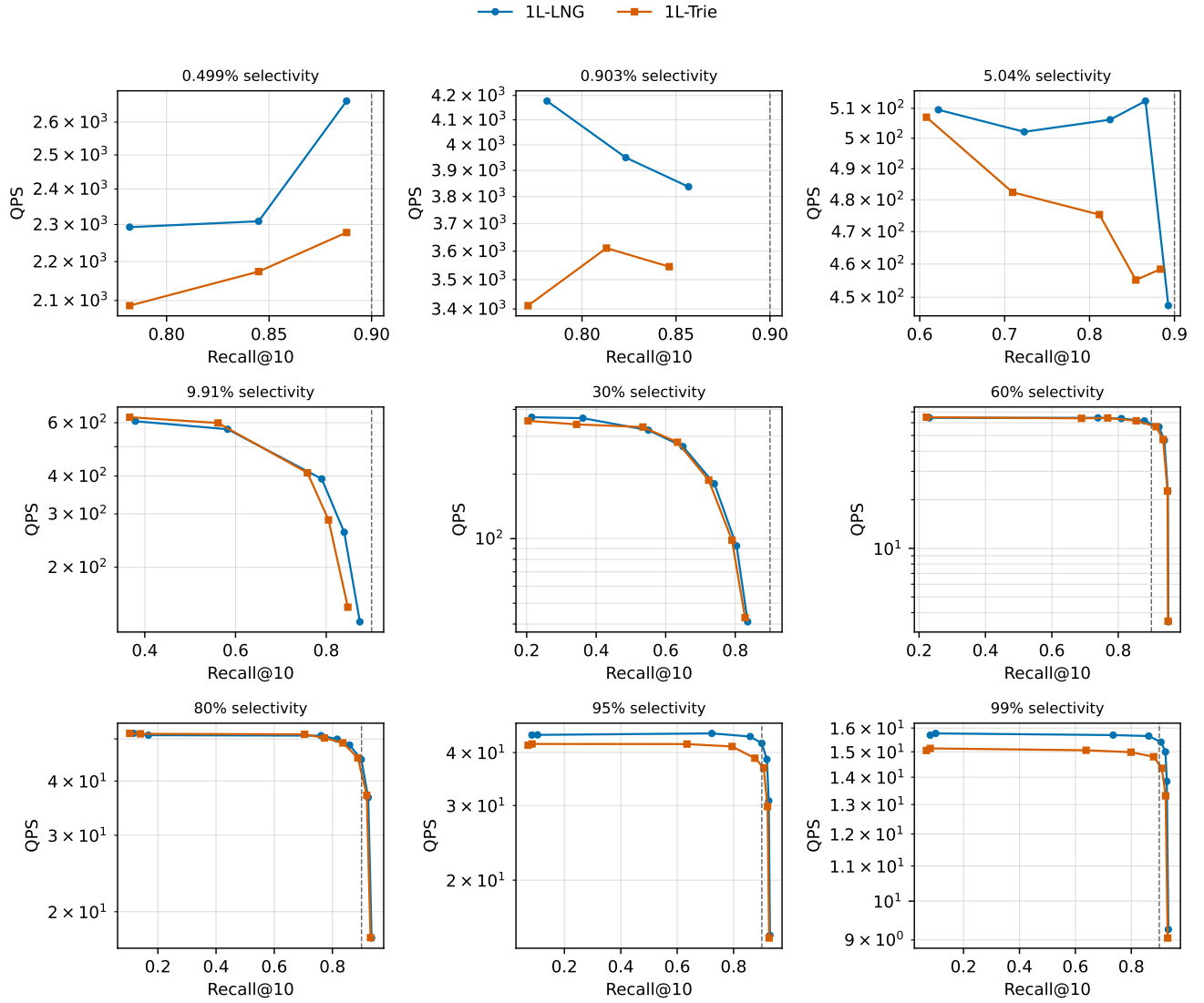


Figure 6: Measured Recall@10-QPS curves for 1L-LNG/1L-Trie overlay topology at fixed threshold, level-0 LNG, and entry provider. Markers are executed points; curves are not interpolated.

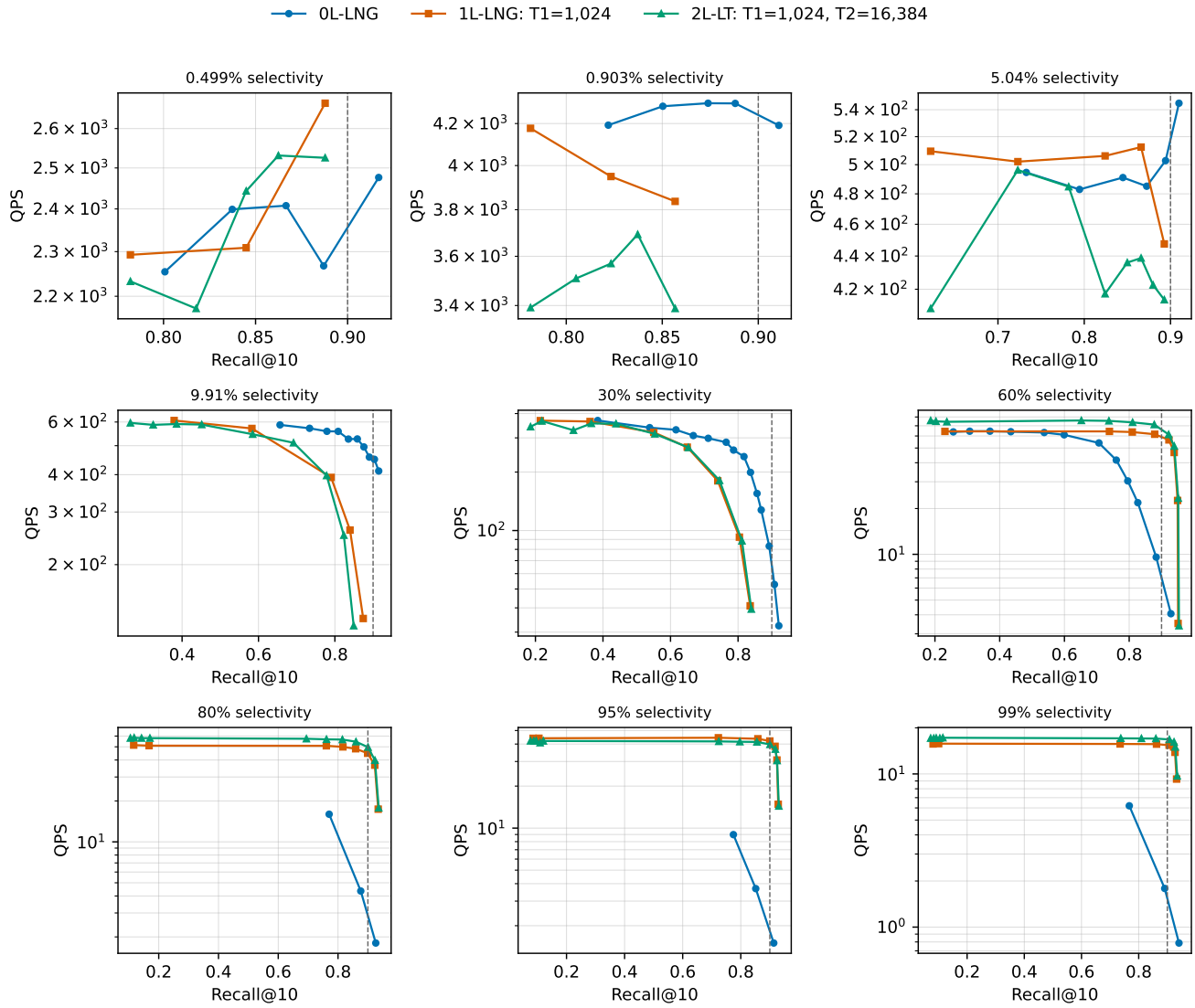


Figure 7: Measured Recall@10-QPS curves for the representative 0L/1L/2L comparison. Markers are executed points; curves are not interpolated.

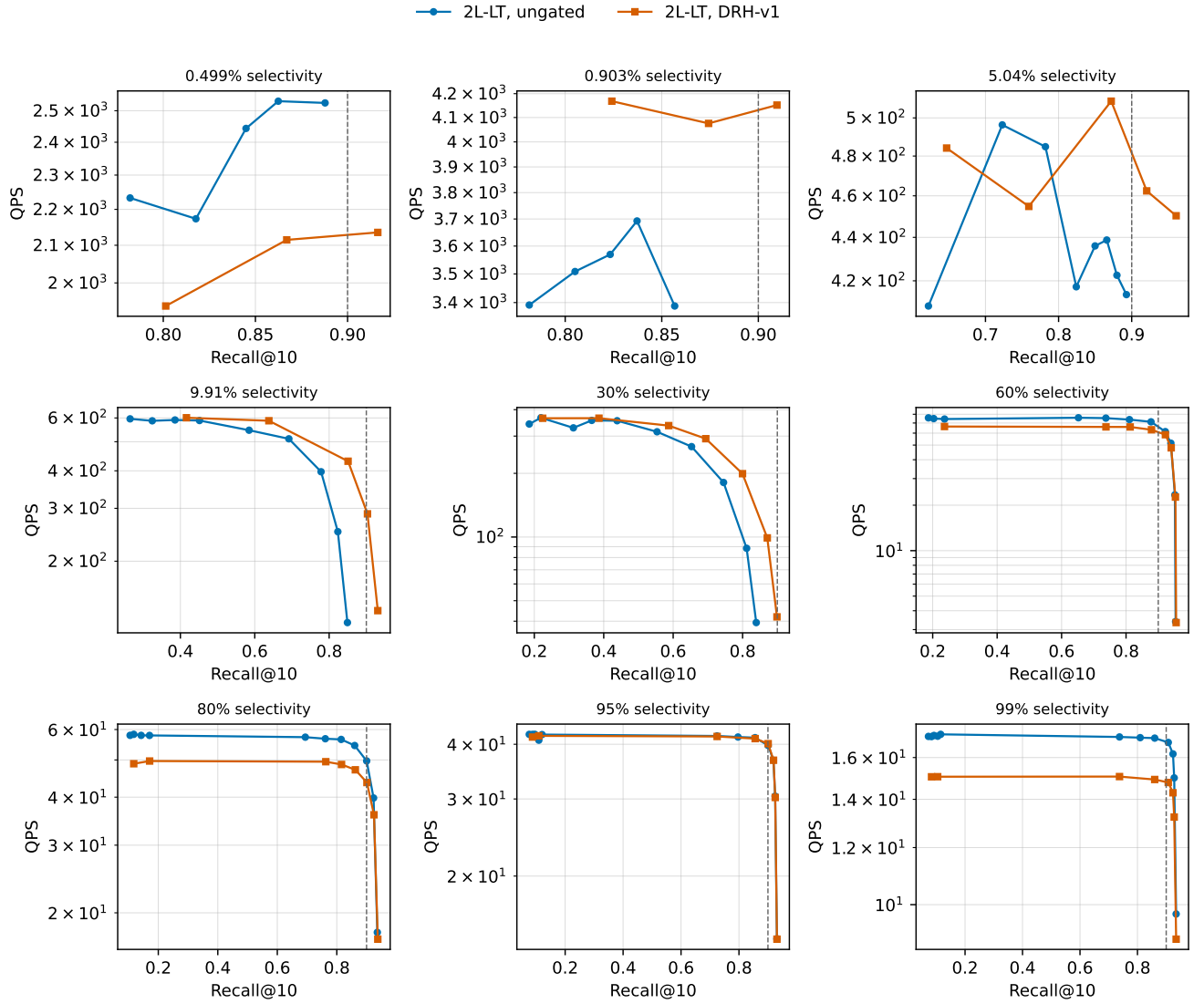


Figure 8: Measured Recall@10–QPS curves for ungated versus DRH-v1 routing for 2L-LT. Markers are executed points; curves are not interpolated.